

Team Name: Elite Coders

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Project Name: Jet Cultivation

Experiment 4: Class Diagram

Procedure: Creating Class Diagram in draw.io and Uploading to Azure DevOps

Step 1: Open draw.io (diagrams.net)

1. Open browser and go to: <https://app.diagrams.net>
2. Choose Device / Google Drive
3. Click Create New Diagram

Step 2: Create New Diagram

1. Enter file name (e.g., *Class Diagram*)
2. Select Blank Diagram
3. Click Create

Step 3: Add Class Shapes

1. From left panel, go to UML → Class
2. Drag class boxes into workspace
3. Create classes:
 - User
 - Farmer
 - Dashboard
 - Crop Journey
 - River
 - Soil
 - Selling Center
 - Irrigation Method
 - Crop

Step 4: Add Attributes and Methods

User

- userId: int
- name: String
- email: String
- password: String
- login(): boolean
- logout(): Boolean

Farmer

- soilType: String
- location: String
- viewDashboard(): void

Dashboard

- dashboardId : int
- displayRiverStatus() : void
- showCrops() : void
- showSoilData() : void
- showSellingCenters() : void

CropJourney

- journeyId : int
- cultivationStage : String
- fertilizationStage : String
- harvestStage : String
- getProgress() : String

River

- riverId : int
- riverName : String
- waterLevel : String
- getRiverStatus() : String

Soil

- soilId : int
- soilType : String
- fertilityLevel : String
- getSuitableCrops() : List

SellingCenter

- centerId : int
- centerName : String
- location : String
- contactNumber : String
- getNearbyCenters() : List

IrrigationMethod

- methodId : int
- methodName : String
- description : String
- suitabilityCondition : String

Crop

- cropId : int
- cropName : String
- growthDuration : String
- requiredSoilType : String
- Step 5: Create Relationships

1. Use Connector lines to link classes

Connections:

- User → Farmer (Inheritance)
- Farmer → Dashboard
- Dashboard → CropJourney
- Dashboard → River
- Dashboard → Soil
- Dashboard → SellingCenter
- Soil → Crop
- River → IrrigationMethod

Step 6: Add Relationship Types

- Use arrow for inheritance
- Use line for association
- Use multiplicity (1, *, etc.) if needed

Step 7: Arrange Diagram

1. Keep User at top
2. Farmer below User
3. Dashboard in center
4. Other classes around it
5. Maintain neat spacing

Step 8: Save Diagram

1. Click File → Save As (.drawio)
2. Export as:
 - PNG / JPG

Step 9: Open Azure DevOps

1. Go to your project
2. Navigate to:
3. Wiki or Repos

Step 10: Upload Diagram

Wiki:

1. Open page
2. Click Edit
3. Insert Image
4. Upload diagram

Repos:

1. Click Upload Files
2. Add image
3. Commit

Step 11: Verify

1. Open uploaded diagram
2. Check clarity and alignment

Result

The Class Diagram for Jet Cultivation System is successfully created using draw.io and uploaded to Azure DevOps.

